INCH-POUND

MIL-D-14951A (AR)
AMENDMENT 4
16 June 1993
SUPERSEDING
AMENDMENT 3
26 August 1992

MILITARY SPECIFICATION

DELAY ELEMENT M53
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-D-14951A (AR) dated 8 September 1982, and is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of Department of Defense.

PAGE 1

2.1.1 Add under Military "MIL-Q-9858 - Quality Program Requirements" cited in 4.1.5

PAGE 2

Add new paragraphs:

"3.1.1 Control environment. Mixing of the Chemical ingredients, loading of the chemical composition and sealing of the M53 delay shall be performed in an environment of 45%-60% RH and 70-80 degrees F."

"3.1.2 Mixing time. The minimum mixing time requirement for the M53 delay mix shall be one (1) hour and thirty (30) minutes."

"3.1.3 Burn time. The bottom (Slow) delay composition shall be varied to achieve the proper burn time. Burn time STD. DEV shall not exceed 0.2 seconds. The top (Fast) delay composition shall have an average burn time of 1 to 1.5 seconds. Burn time STD. DEV shall not exceed 0.2 seconds. The two delay compositions shall be stored in containers of different colors and each container shall be labeled."

FSC 1390

AMSC N/A

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* Paragraph 3.1.3, delete in its entirety and substitute:

"3.1.3 Burn time. The bottom (Slow) delay composition shall be varied to achieve the proper burn time. Burn time STD. DEV. for bottom and top delay composition shall not exceed 0.18 second. The top (Fast) delay composition shall have an average burn time of 1 to 1.5 seconds. For delay batch acceptance at ambient temperature, the average burn time shall be 2.4 +/- 0.1 seconds, and the standard deviation shall be 0.18 second maximum when tested as specified in 4.5.1.3".

* Add new paragraph:

"3.1.3.1 Accelerated aging for sublot. For the accelerated aging test performed at -50 F on one (1) sublot per delay lot, the shift in mean burn time from that at LAT shall be no greater than seven (7) percent when tested as specified in 4.5.2.2."

Add new paragraph:

"3.1.4 Column range. The acceptable column height range value for each delay charge shall be included in the contractor process document."

Add new paragraph:

"3.1.5 Final control. "The delays shall be put into their final pack within five days after lot acceptance. The LAT sample of delays shall be selected after the sub lot of delays has been x-rayed. Sealed M53 delays shall not sit unpacked more than five (5) days in an uncontrolled environment."

Add new paragraph:

* "3.1.6 Arming pin output. The M53 delay assembled and staked into delay holders shall function such that the delay arming pin is expelled, remains intact and does not deform when tested as specified in 4.5.2.1."

* 3.3.2: Add the following to the end of the paragraph, "For sublot acceptance at ambient temperature the standard deviation shall be 0.18 second maximum."

3.8: Delete "shall not exceed 0.50 percent max" and substitute "shall not exceed 0.10 percent max by weight."

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Add new paragraph:

"3.11 Process control. The contractor shall submit a process control document to the Government specifying the entire process for the production of the M53 delay element. The Process Control Document shall be submitted in accordance with paragraph 4.1.5."

Add new paragraph:

"4.1.5 Submission of entire process control document. The contractor shall submit a process control document which details all the specific procedures involved in the entire processing operations. This document shall be a part of the contractor's quality program requirement in accordance with MIL-Q-9858. It shall be submitted to the contracting officer for review by the technical agency. The contractor shall not initiate production without written approval of the entire procedures from the contracting officer. Once approved the contractor shall not change the process without written approval from the contracting officer."

PAGE 6

Delete Table I First article inspection page 2 of 2 and substitute new page 6 included:

PAGE 12

4.4.2.5, Major Defect 104: Delete "Primer below flush surface of holder" and substitute "Depth of primer below surface of holder, max."

PAGE 14

4.4.2.7:

Major 103: Delete the phrase "primer holder cavity or".

Major 104: Delete in its entirety.

4.4.2.8.1:

Critical 1: Change "160 degrees F" to "145 degrees F".

Critical 3: Change "-65 degrees F" to "-50 degrees F".

Major 101: Change "160 degrees F" to "145 degrees F".

Major 103: Change "-65 degrees F" to "-50 degrees F".

Major 104: Change "160 degrees F" to "145 degrees F".

Major 106: Change "-65 degrees F" to "-50 degrees F".

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Table I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
	Delay Element M53		9246248
			NEXT HIGHER ASSEMBLY 9246246

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Delay Element Assembly, M53 (Dwg. 9246248)			
	Examiniation for defects	10	3.2	4.4.2.7
	Arming and delay function test at 145 F.	315	3.3	4.5.2
	Arming and delay functioning test at 70 F.	315	3.3	4.5.2
	Arming and delay functioning test at -50 F.	315	3.3	4.5.2
	Waterproofness test	50	3.4	4.5.3
	Arming pin pull test	25	3.2	4.5.4

NOTES:

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PAGE 16

4.4.2.8.1: In the column marked no. of sample units, add the number "105" for the following defect categories:

Critical 1, 2, 3:

Major 101, 102, 103, 104, 105.

* Paragraph 4.4.2.8.1:

In the column marked category major add "109", in the column examination or test add "arming pin output test", in the column no. of sample units add "80", in the column AQL or 100% add "Acc-1/Rej-2", in the requirement para. add "3.1.6", in the inspection method add "4.5.2.1".

In the column marked category major add "110", in examination add "accelerating aging test for sublot", in no. of sample units add "100", in AQL or 100% add "shift in mean burn time no greater than 7%, in requirement para. add "3.1.3.1", in inspection method and "4.5.2.2".

* Paragraph 4.4.2.8.1:

In the column marked category major add "111", in examination add "hardness", in no. of sample units add "30 per heat treat batch", in AQL or 100% add "Acc-0/Rej-1" in requirement paragraph add "3.2", in inspection method add "Rockwell hardness tester".

PAGE 17

Delete footnote #3 in its entirety and substitute:

"The lot shall be rejected if three (3) or more samples fail to meet the requirement."

PAGES 16 & 17

Delete paragraph 4.4.2.8.1 (pages 16 & 17) in its entirety and substitute new pages 16 and 17 included:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.8.1	Delay Element, M53 (Testing)			9246248
				NEXT HIGHER ASSEMBLY
				9246246

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Delay functioning, short time at 145 F	105		3.3.2	4.5.2
2	Delay functioning, short time at 70 F	105		3.3.2	4.5.2
3	Delay functioning, short time at -50 F	105		3.3.2	4.5.2
4	Arming pin pull test, below minimum load	500	2/	3.2	4.5.4
Major					
101	Arming functioning, non-functioning at 145 F	1/	4/	3.3.1	4.5.2
102	Arming functioning, non-functioning at 70 F	1/	4/	3.3.1	4.5.2
103	Arming functioning, non-functioning at -50 F	1/	4/	3.3.1	4.5.2
104	Delay functioning, above max. time at 145 F	1/	4/	3.3.1	4.5.2
105	Delay functioning, above max. time at 70 F	1/	4/	3.3.1	4.5.2
106	Delay functioning, above max. time at -50 F	1/	4/	3.3.1	4.5.2
107	Arming pin pull test, failure to move above maximum load	500	2/	3.2	4.5.4
108	Waterproofness test	60	3/	3.4	4.5.3

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.8.1	Delay Element, M53 (Testing)				9246248
					NEXT HIGHER ASSEMBLY
					9246246
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AOL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	1/ The number of units samples used in the critical category # 1 to 3 shall be the same as for major category # 101 to 106. Except the criticals shall have no defects permitted. 2/ Compute pull load mean ($\bar{x}$) and the standard deviation (unbiased) (s). Rejection criteria is as follows: a. $\bar{x} - 5s < 22$ b. $\bar{x} + 2.5s > 80$ c. $x \leq 40$ d. Any individual (xi) less than 30. e. Any two (2) individuals (xi) greater than 80. 3/ The lot shall be rejected if three (3) or more samples fail to meet the requirements. 4/ After combining the three (3) phases of the critical category # 1 to 3, the sublot shall be rejected if three (3) or more samples fail to meet the requirements.				

NOTES:

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Add new paragraph:

"4.5.1.2 In process test. The loading machine shall not be altered in any way as a result of the in-process test (i.e. testing five (5) units for time/function every five hundred that are produced). This data should be recorded and used for information only. Corrective action-readjust process to original plan. Record the reason or reasons for the process deviation."

Add new paragraph:

"4.5.1.3 After screening. After screening, test for burning time five (5) samples from each new fast and slow delay mix bucket to evaluate mixture uniformity. Each thirty (30) pound mix yields 5-6 buckets of delay composition. If the burning time standard deviation of any mix does not meet the requirements of paragraph 3.1.3, the composition shall be rejected and ARDEC shall be notified immediately for disposition of the delay composition batch. The delays will be loaded by hand weighing and tamping each increment. Consolidation may be performed by hand or machine loader at the contractor's option. The fast burn mix will be evaluated by loading delays with 244 mg (181 + 63) of that delay composition only. Once uniformity of the faster burning composition is established, the slower burn composition will be loaded as the first increment (181 mg) and the qualified fast burn mix from one bucket utilized as the second increment (63 mg) to load up the test samples of five (5) from each bucket of slow burn composition. The identity of samples to buckets shall be retained and all data forwarded to ARDEC for evaluation."

* 4.5.1.3: Add to 3rd sentence on 5th line of the paragraph after "If the burning time" the following: "average or".

4.5.2, Line 3: Change "160" to "145" and "-65" to "-50".

Add new paragraph:

* "4.5.2.1 Arming pin output test. This test shall consist of firing M53 delay sample assembled and staked into delay holders. The sample delays shall be tested at ambient temperature. Once fired, the delay arming pin in the delay holder shall be inspected. If the requirements of 3.1.6 are not met, the lot shall be rejected."

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Add new paragraph:

* "4.5.2.2 Accelerated aging test for sublot. M53 delay sample shall be condition at 145 F for 72 hours. The delay sample shall then be conditioned at -50F for 24 hours and function tested. Any item failing to meet the requirement of 3.1.3.1 shall result in rejection of the lot.

$$BT\ \% = \frac{\text{Ave BT (final -50 F) - Ave BT (original -50 F)}}{\text{Ave BT (original -50 F)}} \times 100$$

Ave = average

BT = burn time"

PAGE 21

Add new paragraph:

* "6.6 Batch and sublot definition.

1. A batch is approximately thirty (30) pounds of one (1) time blended mix of M53 delay composition.

2. A sublot is approximately one (1) days production of loaded M53 delay element."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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